

SACE User Manual

Secure Adaptive Continuum Explorer (SACE)
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Introduction

The Secure Adaptive Continuum Explorer (SACE) is a soft robotic manipulation system designed for navigating constrained environments and performing adaptive grasping using a tendon-driven continuum arm. Unlike traditional rigid manipulators, SACE leverages compliant structures to achieve safe interaction and high dexterity.

This manual provides a comprehensive guide for end-users to understand the system setup, operation, and safety protocols. It is structured to ensure that even users with minimal robotics experience can successfully operate the system.

- Set up the system
- Operate the robot using teleoperation
- Understand system feedback and safety mechanisms

Intended Users

The SACE system is designed for applications that require flexible, adaptive, and safe robotic manipulation in constrained or hard-to-reach environments. Its continuum architecture makes it particularly suitable for domains where traditional rigid manipulators are ineffective.

- Industrial maintenance personnel performing inspection or repair in confined spaces
- Medical and surgical researchers exploring minimally invasive robotic systems
- Search and rescue professionals requiring flexible robotic tools for navigation and object retrieval

The system is especially useful in scenarios demanding high dexterity, compliance, and safe interaction with uncertain environments.

Prerequisites:

- Basic familiarity with Python execution
- Understanding of USB connections and hardware setup
- No prior robotics experience required for basic operation

System Overview

SACE integrates mechanical, electrical, and software subsystems into a cohesive platform. The continuum arm enables smooth, flexible motion, while the control architecture ensures reliable execution of user commands.

Key Components:

- Continuum robotic arm (4 segments, 10 DOF)
- Tentacle-based soft gripper
- Dual microcontroller system (Master–Slave)
- Laptop-based control interface
- Endoscope camera for visual feedback
- Limit switch safety system

Safety Information

Safe operation is critical due to the presence of moving mechanical components and electrical systems. Users must follow all precautions to prevent damage to the robot and ensure personal safety.

- Do not exceed bending limits ($\sim 30^\circ$ per segment)
- Ensure all wiring is secure before powering ON
- Keep fingers away from moving tendons and motors
- Use Emergency Stop (SPACE key) in case of abnormal motion
- Always start with the arm in a neutral configuration

Hardware Setup

Proper hardware setup ensures reliable operation and prevents system failures. The modular design allows straightforward assembly and debugging.

Mechanical Assembly:

- Tentacle $\rightarrow$ Continuum Links $\rightarrow$ Base Plate $\rightarrow$ Lead Screw $\rightarrow$ Enclosure
- Verify tendon routing and avoid tangling
- Ensure structural rigidity

Electrical Connections:

- Master Pico $\rightarrow$ Laptop (USB)
- Master $\leftrightarrow$ Slave Pico (I2C)
- Motor Drivers $\rightarrow$ Motors (PWM)
- Encoders $\rightarrow$ Slave Pico
- Limit Switches $\rightarrow$ Port Expanders (SPI)

Power Setup:

- Provide 12V input supply
- Ensure 3.3V regulation
- Avoid short between VBUS and 3.3V rail

Software Setup

The software stack bridges user input and hardware execution. It includes host-side Python scripts and embedded firmware on microcontrollers.

Requirements:

```
pip install opencv-python pyserial numpy
```

Steps:

- Flash Master Pico with /src/master code
- Flash Slave Pico with /src/slave code
- Connect devices and identify COM port

Run:

```
cd src/laptop  
python pc_host.py
```

Operating the Robot

The system is controlled through an intuitive teleoperation interface combining keyboard and mouse inputs. Motion is guided using a Follow-The-Leader strategy.

Controls:

- Mouse Click – Add trajectory point
- W / S – Extend / retract lead screw
- U / D – Open / close gripper
- R – Undo last motion
- SPACE – Emergency stop
- Q – Quit

Operation Flow:

- Start system and launch control script
- Use camera feed to guide motion
- Input trajectory points
- Perform grasping and retraction

Feedback and System Behavior

The system continuously monitors safety and performance using sensors and feedback loops. This ensures robust and safe operation even under uncertain conditions.

Feedback Messages:

LIMIT,<mcp_id>,<pin>

TIMEOUT,<motor>,<angle>,<direction>

- LIMIT indicates obstacle or boundary reached
- TIMEOUT indicates unsafe motion duration

Automatic safety mechanisms pause motion and trigger corrective behavior.

Typical Use Workflow

The following workflow outlines a typical interaction scenario:

- Power ON system
- Run control script
- Locate object via camera
- Define trajectory
- Extend, grasp, and retract

Troubleshooting

Common issues can often be resolved through systematic checks of hardware and software components.

- Robot not moving: Check USB, COM port, firmware
- Jerky motion: Inspect tendons and spools
- No camera feed: Verify camera connection
- Sudden stops: Check limit switches

Operational Constraints

The system operates within certain physical and design limitations which must be respected for reliable performance.

- Max bending $\sim 30^\circ$ per link
- Manual teleoperation only
- Backbone hysteresis may occur
- Tendon slack at extreme motion

Shutdown Procedure

Proper shutdown ensures system longevity and prevents damage.

- Stop motion (SPACE)
- Exit program (Q)
- Disconnect power
- Reset arm

Maintenance Guidelines

Routine maintenance is essential to sustain performance and reliability.

- Check tendon tension
- Inspect spool wear
- Verify electrical connections
- Recalibrate if needed

Additional Resources

- Teleoperation: `/src/laptop/README.md`
- PCB: `/pcb/README.md`
- CAD Models: `/3d_models/`
- Videos: `/others/presentation/video/`

Future Improvements

Potential enhancements include improved autonomy, control precision, and system usability.

- Autonomous navigation
- Improved stability
- Closed-loop control
- Enhanced UI

Summary

SACE is a modular soft robotic system demonstrating the capabilities of continuum robotics in real-world applications. It combines flexibility, safety, and intuitive control, making it a powerful platform for research and demonstration.